FORMULA 1 RACE INTELLIGENCE: STEP-BY-STEP PROJECT DOCUMENTATION

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Live Web App: https://jiyansh-gokhru.github.io/f1-race-intelligence-dashboard/ | **Tableau Public Story:** FORMULA1RaceIntelligenceDashboard/Story1

1. Executive Overview & Multi-Sheet Relational Data Architecture

This project delivers an end-to-end visual analytics and race intelligence platform built on the official 5-sheet **F1 Racing Project Dataset.xlsx** workbook spanning 17 Formula 1 seasons (**2010–2026**) and **13,748 total records**. It integrates a 9-visualization interactive dashboard, a 9-scene guided storyboard titled *Formula 1 Race Intelligence: From Global Overview to Championship Success*, 3 global filter controls, and a responsive Flask / GitHub Pages web application.

Sheet Name	Records & Columns	Primary Join Keys	Core Analytical Fields
1. race_results	6,464 rows × 42 cols	race_id, circuit_id, driver	season (2010–2026), team (10), driver (19), points, finish_position (1,002 podiums), starting_compound, dnf (396), dnf_reason (6)
2. lap_intelligence	3,814 rows × 14 cols	race_id, circuit_id, driver	lap_number, lap_time_sec, sector_1_sec, sector_2_sec, sector_3_sec, tyre_compound, tyre_age_laps
3. qualifying_results	3,130 rows × 16 cols	race_id, circuit_id, driver	q1_time_sec, q2_time_sec, q3_time_sec, best_qual_time_sec, qual_position, pole_position
4. championship_standings	311 rows × 10 cols	season, driver, team	driver_position, driver, team, points, wins, podiums, pole_positions, dnfs, champion
5. circuits	29 rows × 9 cols	circuit_id	circuit_name (29 showcased, 24 active), country (21 nations), city, lap_km, laps, altitude_m

2. Complete Inventory of the 9 Visualizations & Verified Metrics

#	Visualization Title & Chart Type	Tableau Shelf Mapping & Verified Dataset Figures
1	**Executive KPI Cards (4 Cards)**	Competing Teams: **10** | Seasons in Play: **17** (2010–2026) | Race Outcomes Logged: **6,464** | Circuits Showcased: **29** (24 active across 21 nations).
2	**Global Driver Distribution Across GP Nations** (Filled Map)	Country role on Detail, CNT(race_results) on Color. Bahrain leads (**2,168** / 33.54%), followed by USA (**400**), Monaco/Hungary/Spain (**328** each), Canada/UAE/Brazil (**308** each).
3	**Driver Reliability Analysis** (Stacked Bar Chart)	**396 total DNFs** (6.13% rate) across 6 causes: Engine (**73**), Hydraulics (**70**), Mechanical (**67**), Brake failure (**64**), Gearbox (**62**), Collision (**60**). Highest DNFs: Sainz (**34**), Perez (**29**), Leclerc (**28**). Lowest front-runner DNFs: Hamilton (**12**).
4	**Driver Performance Leaderboard** (Vertical Bar + Top N)	Controlled by **Top N Parameter** (default 10). Direct race_results points: Verstappen (**4,640**), Hamilton (**4,491**), Perez (**3,366**), Russell (**3,299**), Leclerc (**3,243**). Logical Join mode: Hamilton (**31,147**), Verstappen (**31,126**), Leclerc (**27,673**), Norris (**26,408**).
5	**Team Performance Ranking** (Horizontal Bar Chart)	Red Bull Racing (**8,006** pts), Mercedes (**7,790** pts), Ferrari (**5,778** pts), McLaren (**4,270** pts), Alpine (**1,947**), Aston Martin (**1,181**), Williams (**1,057**), RB (**923**), Haas (**591**), Kick Sauber (**363**).
6	**Team Championship Performance Analysis** (Vertical Bar)	From championship_standings: Red Bull Racing (**8,006** pts, 100 wins, 7 titles), Mercedes (**7,790** pts, 118 wins, 7 titles), Ferrari (**6,390** pts, 46 wins), McLaren (**4,270** pts, 35 wins, 3 titles).
7	**Tyre Compound Usage Distribution** (Pie Chart)	Starting Compound across 6,464 starts: Soft (**1,889** / 29.22%), Medium (**1,884** / 29.15%), Hard (**1,849** / 28.60%), Intermediate (**842** / 13.03%).
8	**Country-wise Race Participation** (Treemap)	Proportional rectangle sizing across 21 countries. Top 5 host nations (Bahrain, USA, Monaco, Hungary, Spain) represent **54.95%** (3,552 / 6,464) of all entries.
9	**Podium Achievements by Driver & Team** (Packed Bubble)	**1,002 total podiums**: Verstappen (**187**), Hamilton (**174**), Russell (**115**), Leclerc (**113**), Perez (**97**), Norris (**85**), Sainz (**63**: 44 Ferrari, 19 Williams), Piastri (**35**), Ocon (**25**).

3. Calculated Fields, Global Filters & 9-Scene Tableau Storyboard

Calculated Fields & Parameters: (1) **DNF Count**: `IF [Dnf] = 1 THEN 1 ELSE 0 END` (396 total DNFs); (2) **Points Cumulative**: `RUNNING_SUM(SUM([Points]))`; (3) **Podium Count**: `IF [Finish Position] <= 3 THEN 1 ELSE 0 END` (1,002 podiums); (4) **Top N Parameter**: Integer parameter (Range 3–19, Default 10) paired with `INDEX() <= [Top N Parameter]`.
3 Global Dashboard Filters: (1) **Dnf Reason Filter** (6 failure modes); (2) **Year of Race Date / Season Slider** (2010–2026 range); (3) **Top N Parameter Control**.

Scene	Navigator Caption	Guided Narrative & Key Analytical Insight
Scene 1	**F1 Overview KPIs**	Establishes project scope: 10 constructors, 17 seasons (2010–2026), 6,464 race starts, and 29 circuits.
Scene 2	**Geographic Distribution**	Maps 21 host countries; highlights Bahrain (2,168 entries) and multi-circuit USA expansion (400 entries).
Scene 3	**Driver Reliability Analysis**	Breaks down 396 DNFs across Engine (73), Hydraulics (70), Mechanical (67), Brake failure (64), Gearbox (62), Collision (60).
Scene 4	**Driver Leaderboard**	Compares Verstappen (4,640 pts) and Hamilton (4,491 pts) in direct race_results and 31,147 vs 31,126 in Logical Join mode.
Scene 5	**Team Performance Ranking**	Ranks Red Bull Racing (8,006 pts) and Mercedes (7,790 pts) at the top of constructor race scoring.
Scene 6	**Team Championship Analysis**	Contrasts Red Bull (7 titles, 100 wins), Mercedes (7 titles, 118 wins), Ferrari (6,390 standing pts), and McLaren (3 titles).
Scene 7	**Tyre Strategy Distribution**	Reveals near-equal dry compound usage (~29% each for Soft, Medium, Hard) plus 13.03% Intermediate starts.
Scene 8	**Country Participation**	Demonstrates via Treemap that the top 5 host countries account for 54.95% of all logged race participations.
Scene 9	**Podium Achievements**	Visualizes 1,002 podiums via Packed Bubbles, led by Verstappen (187), Hamilton (174), Russell (115), and Leclerc (113).

4. Performance Testing, Responsive Design & Flask / GitHub Pages Integration

All 14 MySkillWallet project milestones were verified against strict performance and visual benchmarks:

Audit Dimension	Target Requirement	Measured Result & Verification Status
Data Accuracy & Integrity	100% match with 5-sheet Excel	**PASS** — 13,748 rows validated via Python ETL; zero null keys or duplicate race_id entries.
Dashboard Load & Filter Latency	< 2.0 seconds load time	**PASS** — 0.42s initial render; < 45ms interactive filter/slider response across all 9 charts.
Cross-Device Responsiveness	Desktop (1920px) to Mobile (375px)	**PASS** — Zero horizontal overflow; responsive CSS grid + Tableau JavaScript Embed API v3.
Web Application Deployment	Flask + Live GitHub Pages URL	**PASS** — Flask routes (/ , /api/summary, /api/telemetry, /health) + live static build deployed.